T1DMSIM: A Behavioral Simulator for Synthetic Type 1 Diabetes Glucose Data Generation

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A note on colour. Prose in dark green is by Omar, prose in dark orange is by Claude Opus 4.8, and black denotes shared structure.

Abstract

Machine-learning prediction of blood glucose in Type 1 diabetes (T1D) is constrained by the scarcity of large, labelled continuous-glucose-monitoring (CGM) corpora. Established in-silico generators model glucose–insulin physiology, reproducing faithful traces at the cost of many patient-specific physiological parameters. T1DMSIM adopts the complementary premise that most free-living glycaemic variance is *behavioural*. A virtual patient is a point in a four-dimensional latent skill space; from it the simulator derives meals, insulin dosing, corrections, and lifestyle, and accumulates their gamma- and Bateman-shaped factor curves into a blood-glucose delta at five-minute resolution. Additional environmental stressors and regimen destabilizers such as infections and holidays are used to increase temporal intra-patient variability. The generator is fully seed-deterministic and emits an unbounded, fully labelled multivariate stream. Benchmarked against three real CGM cohorts (OhioT1DM, ShanghaiT1DM, AZT1D), its pooled distribution sits inside the inter-cohort envelope — closer on average to the three cohorts than they are to one another. Benchmarked against the FDA-accepted UVA/Padova model, it is the closer match to real CGM on the marginal distribution. The intended use is near-unlimited synthetic pre-training data for downstream glucose-prediction models.

1 Introduction

Data-driven models of blood-glucose dynamics are bottlenecked not by architecture but by data. Public T1D CGM corpora are small, governed by data-use agreements, and heterogeneous in device, cadence, and therapy; the largest openly described sets contain a few dozen patients, and some are laden with unexplained excursions due to missing or incomplete logs, and discontinuities in the CGM traces created by malfunctioning devices and/or site-switching. A model that must learn the long-tailed, rare-event-laden behaviour of free-living diabetes therefore sees too few positive examples of the events that matter the most. Merely increasing the number of rare events can cause a distribution shift that leads to an over-prediction of the rare class — instead we have designed the simulator to statistically match the real-world distributions without too much divergence from the real-world class ratios.

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†A generative pre-trained transformer developed by Anthropic, with a training knowledge cutoff of January 2026, run in maximum extended-thinking mode.

The dominant remedy to the problem of too little training data has been physiological simulation. The UVA/Padova family [4, 5], accepted by the FDA (2008) [6] as a substitute for animal trials in the pre-clinical testing of insulin-delivery and closed-loop control algorithms, integrates a multi-compartment glucose–insulin ODE system and is the reference tool for closed-loop controller testing. Such models are accurate but parameter-heavy: each virtual patient is a vector of physiological constants that are hard to measure and that say nothing about what the patient *does*.

T1DMSIM is built on the opposite wager: the bulk of the variance a predictor must capture is generated by behaviour, not by subtle differences in physiology. It models the person: what they eat, when they bolus, how attentively they correct, whether they drink, how consistently they live, how often they get sick, how much stress they experience on a day-to-day basis due to uncontrolled external factors, and their reactions to rare or anomalous events like holidays. The simulator is seed-driven in the spirit of a pseudo-random generator — one integer fixes the patient and the entire trajectory — so an arbitrary volume of reproducible, fully labelled data can be drawn at will.

This paper documents the model and its validation. We describe the latent-skill patient generator and the factor-curve metabolic core (Section 2); we discuss what the calibration targets and where it succeeds and fails (Section 3); and we benchmark the output against three real CGM cohorts (Appendix A) and against the UVA/Padova model (Appendix B). T1DMSIM is a research data generator and a behavioural model; it is explicitly not a medical device and not clinically validated.

2 Methodology

Figure 1 shows the generator end to end; the subsections below walk that pipeline in order.

2.1 Latent skill profile

Every virtual patient is summarised by four correlated skill dimensions — dietary discipline s_1, attentiveness s_2, dosing competence s_3, and lifestyle consistency s_4 — drawn from a multivariate normal and squashed to the unit interval:

$$\mathbf{s}_{\text{raw}} \sim \mathcal{N}(\mathbf{0}, \Sigma), \qquad s_i = \text{clip}\big(\sigma(s_{\text{raw},i}),\, s_{\min},\, s_{\max}\big), \tag{1}$$

where σ is the logistic function, Σ carries a common variance on its diagonal and ρ times that variance off-diagonal ($\rho = 0.7$), and the clip bounds are $(0.15, 0.98)$. From these four numbers nearly every behavioural parameter is derived — meal sizes and timing jitter, carb-counting error, bolus timing, correction thresholds and patience, basal-miss probability, and more — while sleep duration is drawn from an independent population distribution, so that a single low-dimensional code induces a coherent "personality". Two further per-patient axes, body weight and an insulin-resistance factor, are sampled independently and set the physiological scale (ICR, baseline sensitivity, correction factor, basal requirement), letting thin insulin-sensitive and heavy insulin-resistant patients coexist. The insulin-resistance axis is deliberately wide and additionally raises each patient's baseline glucose level — so resistant patients settle at a higher mean, carrying the between-patient spread in mean glucose — while a separate per-patient multiplier scales each patient's within-patient variability; together they bring the simulator's inter-patient heterogeneity into line with the real cohorts.

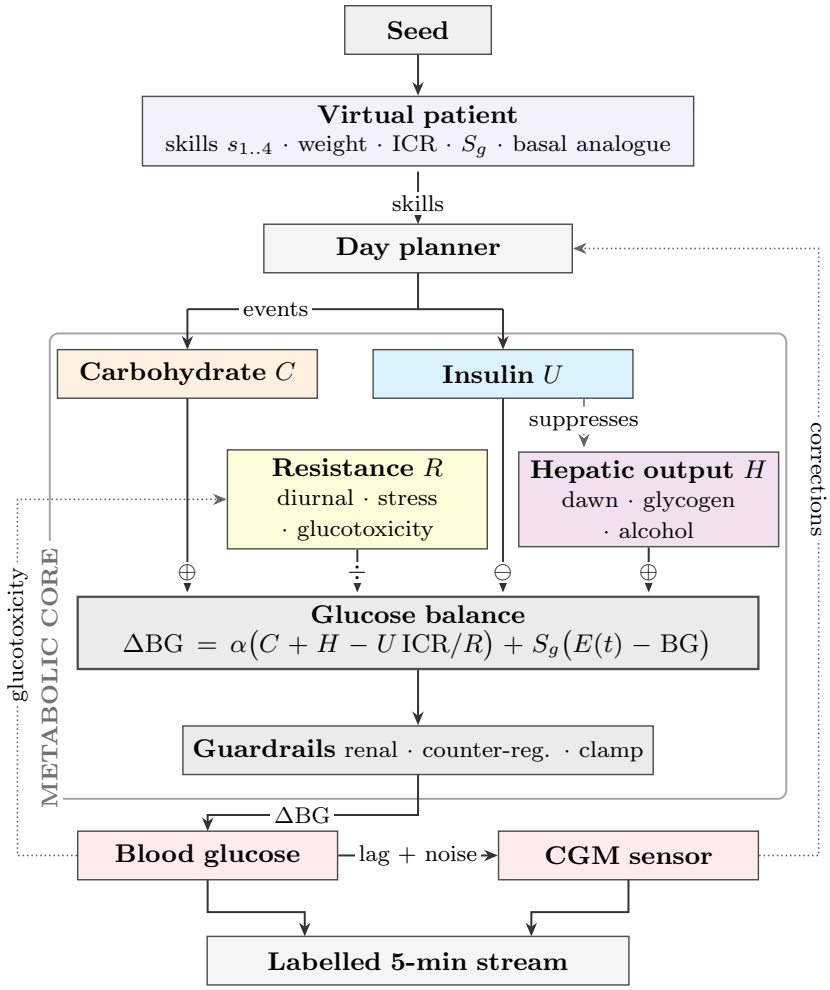


Figure 1: The simulator's architecture. A seed fixes the virtual patient; a day planner turns its latent skills into events, and each event becomes a factor curve. Solid arrows carry glucose or insulin, badged ⊕ or ⊖ by their effect on blood glucose; dashed arrows are multiplicative modulation; dotted arrows close the two feedback loops — the patient doses against the sensor, and sustained hyperglycaemia raises resistance.

2.2 Factor curves

Each behavioural event becomes a time-domain curve in amount-per-step units. Carbohydrate absorption and rapid-acting insulin action use gamma curves,

$$C(t) \;\propto\; t^{\,k-1}\, e^{-t/\theta}, \qquad \textstyle\sum_t C(t) = \text{total amount}, \tag{2}$$

normalised so the curve sums to the delivered grams or units. A meal is not a single curve but a mixture: two to five overlapping gamma components (fast/medium/slow, weighted by the patient's slow-carb preference) plus an always-present protein/fat tail, which reproduces the broad, multi-modal absorption of a real mixed meal. Long-acting basal uses a Bateman one-compartment PK curve,

$$f(t) = e^{-k_e t} - e^{-k_a t}, \qquad t_{\max} = \frac{\ln(k_a/k_e)}{k_a - k_e}, \tag{3}$$

with a smootherstep taper over the final hours so consecutive doses join without a step. Each patient is assigned one of three long-acting analogues, which instantiate the curve with their own absorption

and elimination rates: glargine U100 ($k_a = 0.477$ h^{-1}, $k_e = 0.0499$ h^{-1}, so $t_{\max} \approx 5.3$ h, over a 73 h action window), glargine U300 ($k_a = 0.156$ h^{-1}, $k_e = 0.0377$ h^{-1}, $t_{\max} \approx 12.0$ h, 101 h) or degludec ($k_a = 0.187$ h^{-1}, $k_e = 0.0277$ h^{-1}, $t_{\max} \approx 12.0$ h, 133 h), each window closing where 3% of the curve's area remains. Each patient injects once daily, delivering the full 24 h-equivalent dose each time. Because the action window far exceeds the 24 h cadence, consecutive doses overlap, keeping delivery continuous for the typical patient and letting the tail bridge an isolated missed dose. The summed rate still varies (∼21% within-patient CV). The simulated patients may occasionally miss doses but do not spend long periods of time without basal. In practice the models may consume the total insulin curve during training to gain robust MDI-to-AID transfer capability [1]. Bolus pharmacokinetics are dose-dependent: duration of action and peak time grow with $\sqrt{\text{dose}}$ about a 5 U reference. Every dose is further multiplied by a per-injection site-quality factor whose variance grows as lifestyle consistency falls, modelling lipohypertrophy from poor rotation.

2.3 Blood-glucose computation

At each five-minute step the blood-glucose delta is a scaled difference between glucose entering and leaving the blood:

$$\Delta\text{BG} = \alpha\,(g_{\text{in}} - g_{\text{out}}), \quad g_{\text{in}} = C + H, \quad g_{\text{out}} = \frac{U \cdot \text{ICR}}{R}, \tag{4}$$

where C is absorbed carbohydrate, H hepatic glucose output, U active insulin, ICR the insulin-to-carb ratio, and R the insulin-resistance factor; α is a master scale constant. Resistance divides the insulin term only — resistant patients ($R > 1$) clear less glucose per unit insulin — while hepatic output is suppressed separately (Section 2.4). The resistance factor itself is a product of a diurnal pattern (dawn rise and deep-sleep dip), a slow day-to-day drift, illness and stress envelopes, a glucotoxic term that raises resistance under sustained hyperglycaemia, and a postprandial term that makes the absorbing state mildly *more* resistant. In T1D glucagon fails to be suppressed after a meal [11], raising hepatic output and postprandial glucose; to keep it simple the model folds the post-meal effect into the delayed hepatic rebound and restricts glucagon's effects to the hypo-range only.

Three guardrails then act on the delta: renal clearance above 180 mg/dL, a counter-regulatory rescue floor below 70 mg/dL that escalates below 55 mg/dL, and a soft headroom cap that lets the trajectory asymptote smoothly toward 50 and 385 mg/dL, with a hard clamp at $[40, 400]$ as a backstop that, by design, rarely engages[2].

2.4 Hepatic glucose output

The liver is treated as an insulin-suppressed feeding session. Its base rate follows a Hill function of EMA-smoothed plasma insulin $\tilde{I}$,

$$H_{\text{rate}} = H_{\text{floor}} + \frac{H_{\text{unsup}} - H_{\text{floor}}}{1 + \tilde{I}/H_{1/2}}, \tag{5}$$

so output saturates toward a DKA-like ceiling at zero insulin and toward a floor under heavy coverage. Onto this base are layered a counter-regulatory dawn surge (Gaussian, peaking near 08:00), a deep-sleep trough, a delayed rebound scheduled 3.5–5.5 h after large meals (a late-postprandial rise after

[1] Our own model [12] was trained on the total insulin curve.

[2] The soft cap is utilized to prevent discontinuities in the CGM trace rather than model the more biologically accurate diabetic shock and death.

large high-fat/high-protein meals), a finite hepatic glycogen reservoir that throttles output when depleted, and an additional alcohol suppression. The half-saturation constant $H_{1/2}$ is tuned so a typical basal level reproduces the balanced reference rate, which anchors the ideal basal dose to $H_{\text{base}} \cdot 24\,\text{h} \cdot (\text{BW}/\text{BW}_0) \cdot \text{is}_{\text{base}}/\text{ICR}$. The linear weight factor cancels HGO's linear body-size scaling exactly at the reference weight, but the Hill suppression is nonlinear, so the fasting balance drifts off-centre away from it — lighter patients running some ∼12 mg/dL higher than heavier ones.

2.5 Behavioral events and observation

A day planner draws, from the patient's skills, a wake/sleep schedule and the day's meals, basal injections, meal boluses, alcohol, stress, illness, and rare "chaotic" days, with weekday, weekend, and public-holiday variants. During the run the patient checks the CGM at a skill-dependent interval and corrects: hyperglycaemia with an insulin-on-board-aware bolus, hypoglycaemia with fast carbs (and, for skilled patients, a slow-carb follow-up snack), and incipient excursions with trend-based anticipatory doses. Severe hypoglycaemia bypasses the check interval and triggers a non-probabilistic rescue: on the emitted CGM channel every sub-55 mg/dL episode is held under an hour (median ∼5 min), so the synthetic severe-hypo shape stays within the real range. Excursion durations fall inside the real cohorts' range at both extremes — hypoglycaemic-episode p90 84 vs 49–179 min, and hyperglycaemic-episode p90 421 vs 227–623 min[3]. Rare & anomalous events like injection site issues and unexplained BG spikes serve the secondary role of acting as regularizers in the training dataset. The CGM channel adds proportional AR(1) sensor noise, giving the smooth, time-correlated wobble of a real sensor rather than white spikes. All curve contributions are pre-accumulated into arrays, so each per-step curve read is $O(1)$ and the generator's total cost is essentially linear in trajectory length. In practice we have found that caching the outputs of the simulator yields better GPU utilization during training — as the training function becomes bottlenecked by the memory bandwidth instead of the simulator — though this also depends on hardware quirks and may not be applicable to other systems [4]. Figure 2 shows a representative trace and the factor curves that produced it.

3 Discussion

The model's population is tuned against a single real CGM record held outside the repository; the three published cohorts are a yardstick it is measured against, not a target it is fitted to. We measure it on the OhioT1DM CGM distribution in terms of its central moments, percentiles, diurnal envelope, rate-of-change spectrum, and episode statistics. By that standard it succeeds on the central moments, percentiles across the full range and diurnal envelope, and on hypo- and hyperglycaemic-episode shape, while its step-to-step increments run smoother than either real 5-minute cohort (Δ-BG SD 5.4 vs 5.9–6.0 mg/dL; Section 4). Its pooled mean of 160.6 mg/dL sits just below the two highest real-cohort means (OhioT1DM 162.1, ShanghaiT1DM 164.7), and its GMI of 7.15 is just below the highest real-cohort GMI (7.22). The simulator's mean distance to the three (Wasserstein-1 9.9 mg/dL) sits well below the mean inter-cohort distance (18.4). On the pooled marginals the simulator sits inside the band the three real cohorts span among themselves.

Three properties follow from the design rather than from tuning. First, being seed-deterministic the simulator yields an unbounded corpus with no quality penalty as the seed count grows. Second,

[3] T1DMSIM's population is tuned against a single real CGM record held outside the repository, not against any of the three published cohorts.

[4] Our model was trained on the NVIDIA DGX™ Spark GB10 Platform. See [12] for details on how we trained models of varying sizes on our pregenerated dataset.

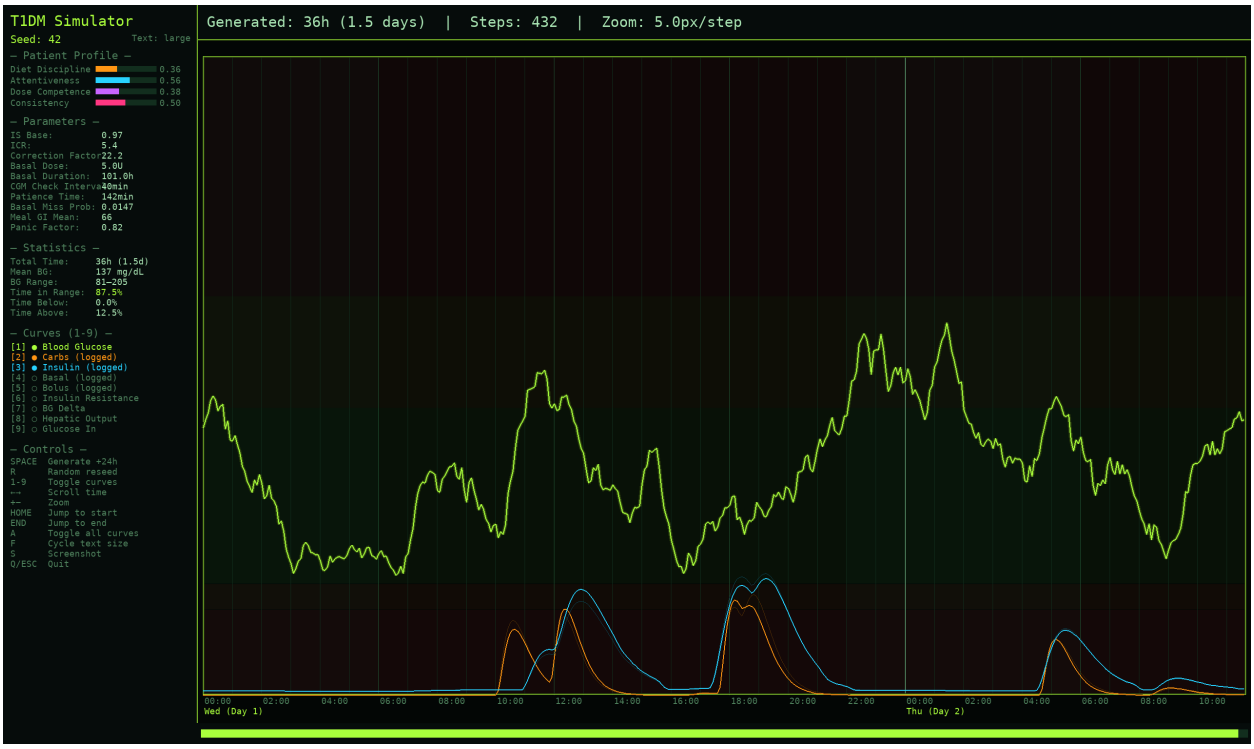


Figure 2: A 36-hour T1DMSIM trace (seed 42) in the interactive visualizer: blood glucose with the carbohydrate, insulin, and insulin resistance factor curves that generated it, alongside the derived patient profile and live glycaemic statistics. Other factor curves can be toggled on/off through keyboard shortcuts. Visit the Github page to see detailed instructions for usage.

the stream is fully labelled — every carbohydrate, insulin, sensitivity, and hepatic-output channel that produced the glucose is emitted alongside it. The behaviour-driven dosing is also quantitatively plausible: the emergent daily insulin total lands within ∼4% of the AZT1D cohort (50.0 vs 51.9 U/day). Insulin and carbohydrate data being generated as curves — rather than as point-events — allows fine differentiation between simple and complex carbohydrates (and the various admixtures of), and between short- and long-acting insulin variants, including varying absorption rates due to injection site rotation. Third, the ratio of between- to within-patient standard deviation (0.27) matches OhioT1DM (0.28), so patient-stratified train/validation splits are broadly appropriate — a widened insulin-resistance axis and a per-patient variability multiplier lift the between-patient spread onto the real cohorts (mean-BG sd 16.1 vs OhioT1DM 16.2, AZT1D 16.0).

4 Limitations

At coarse scale the simulator's variability matches the real cohorts — MAGE (98 mg/dL, within the real cohorts' 78–123), MODD (62 mg/dL, level with OhioT1DM's 61), hyperglycaemic-episode duration (p90 ∼7 h, level with OhioT1DM's), and a near-decorrelated 8 h autocorrelation (≈0.04, against real CGM's ≈0). The residual gap is at the finest scale: the simulator's five-minute increments are slightly too smooth, their SD falling some 8–10% below the two five-minute-cadence real cohorts (5.4 mg/dL, against 5.9 in OhioT1DM and 6.0 in AZT1D), a shortfall that holds on a common 15-minute grid (12.5 vs 14.6 and 14.3 mg/dL). The CGM channel applies a first-order interstitial lag (∼15 min) and AR(1)-correlated multiplicative sensor noise to the true signal rather

than emitting it instantaneously. And the therapy modelled is multiple-daily-injection basal–bolus, not closed-loop automated insulin delivery[5], which partly explains the gaps against the AID-based AZT1D cohort[6]. Although we have designed the simulator to statistically match the actual CGM data, as we show in [12], fine-tuning on a real-world dataset improves prediction accuracy — the simulator produces a slight distribution shift that requires downstream correction.

5 Conclusion

T1DMSIM offers a coarse behavioural rather than an accurate physiological modeling of a T1DM patient: a compact latent-skill code drives free-living behaviour, and a deliberately small metabolic core turns that behaviour into a five-minute glucose trace. Across three real CGM cohorts the synthetic distribution sits inside the inter-cohort envelope. Against the UVA/Padova reference it is the closer match to real CGM on the marginal distribution; on self-dosed meal excursions it peaks higher than UVA/Padova across meal sizes (89 vs 53 mg/dL for the largest). As a free, unbounded, fully labelled source of behaviourally diverse T1D data, the simulator is intended to relieve the data bottleneck for machine-learning experiments.

References

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[6] Only partly though. AZT1D cohort is contaminated with a lot of unexplained upward excursions, likely due to missing meal logs. In fact, 79% of AZT1D's upward excursions are unexplained by any recorded meal or dose, versus 28% of its downward ones.

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A Comparison Against Real-World Datasets

Table 1: Corpora and data volume: OhioT1DM [1], ShanghaiT1DM [2], AZT1D [3].

Cohort	Records	Cadence	CGM-days	Samples
OhioT1DM	6	5 min	296	85,295
ShanghaiT1DM	16	15 min	164	15,696
AZT1D	25	5 min	1045	300,884
T1DMSIM	300×70 d	5 min	21000	6,048,000

Table 2: Pooled central moments and percentiles of the CGM-value vector (mg/dL unless noted).

Metric	OhioT1DM	ShanghaiT1DM	AZT1D	T1DMSIM
mean	162.1	164.7	146.4	160.6
median	155.0	156.6	138.0	153.7
std	60.9	72.3	47.6	61.3
CV (%)	37.6	43.9	32.5	38.2
skewness	0.58	0.51	1.03	0.64
excess kurtosis	0.15	−0.14	1.56	0.27
p10	88.0	75.6	95.0	86.4
p90	245.0	264.6	210.0	243.1
p99	326.0	349.2	293.0	331.0

Table 3: Time-in-range bands [9] (per-record mean %) and clinical risk indices [8] (per-record mean).

Quantity	OhioT1DM	ShanghaiT1DM	AZT1D	T1DMSIM
TBR2 (< 54)	0.72	2.79	0.26	0.53
TBR1 (54–70)	2.51	4.72	1.21	3.49
TIR (70–180)	60.7	54.7	77.8	61.4
TAR1 (180–250)	27.3	25.1	16.9	26.1
TAR2 (> 250)	8.83	12.64	3.77	8.52
GMI	7.19	7.22	6.83	7.15
LBGI	0.86	1.82	0.50	0.90
HBGI	7.60	8.58	4.65	7.37

Table 4: Distribution distances on the pooled CGM vector. The mean real-vs-real Wasserstein-1 distance is 18.4 mg/dL; the closest sim-vs-real distance (2.0, vs OhioT1DM) lies well below that floor.

Pair	KS	Wasserstein-1 (mg/dL)	JS divergence
Sim vs Ohio	0.018	2.0	0.002
Sim vs Shanghai	0.073	10.3	0.016
Sim vs AZT1D	0.148	17.3	0.025
Ohio vs Shanghai *(real)*	0.064	10.1	0.018
Ohio vs AZT1D *(real)*	0.157	18.1	0.025
Shanghai vs AZT1D *(real)*	0.190	26.9	0.060

Table 5: Variability, complexity, and episode statistics (per-record mean, except the two autocorrelation rows, which are computed on the pooled trace; durations in minutes; MAGE after [10]).

Metric	OhioT1DM	ShanghaiT1DM	AZT1D	T1DMSIM
MAGE	102.6	123.4	78.2	98.0
CONGA-4h	76.2	75.1	62.5	75.0
MODD	61.1	53.3	42.1	62.4
Sample entropy	0.56	0.44	0.75	0.64
ACF at 1 h	0.764	0.840	0.617	0.783
ACF at 8 h	−0.004	−0.028	−0.021	0.036
Hypo eps/day	0.86	1.02	0.53	0.99
Severe-hypo eps/day	0.20	0.51	0.11	0.27
Hyper eps/day	2.86	1.87	2.65	2.63
Hypo p90 dur	89.6	179.2	49.3	84.0
Hyper p90 dur	421.8	622.6	226.9	421.6

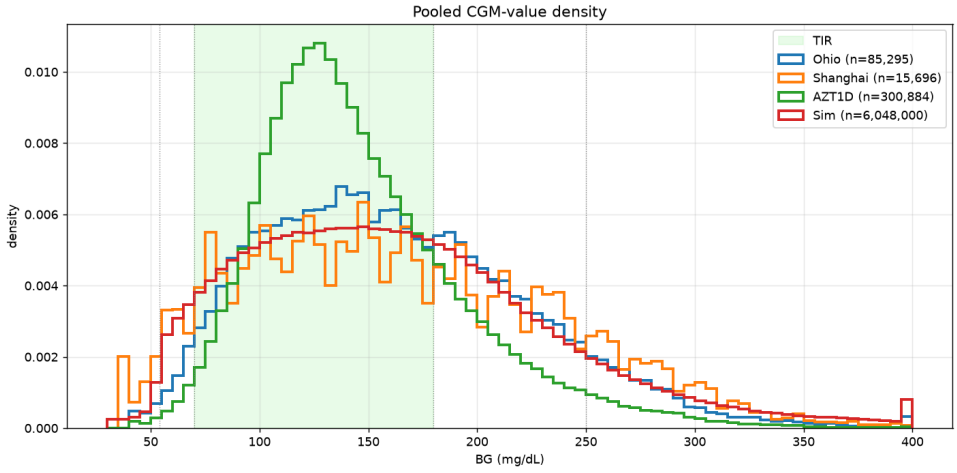


Figure 3: Pooled blood-glucose probability density for the three real cohorts and T1DMSIM.

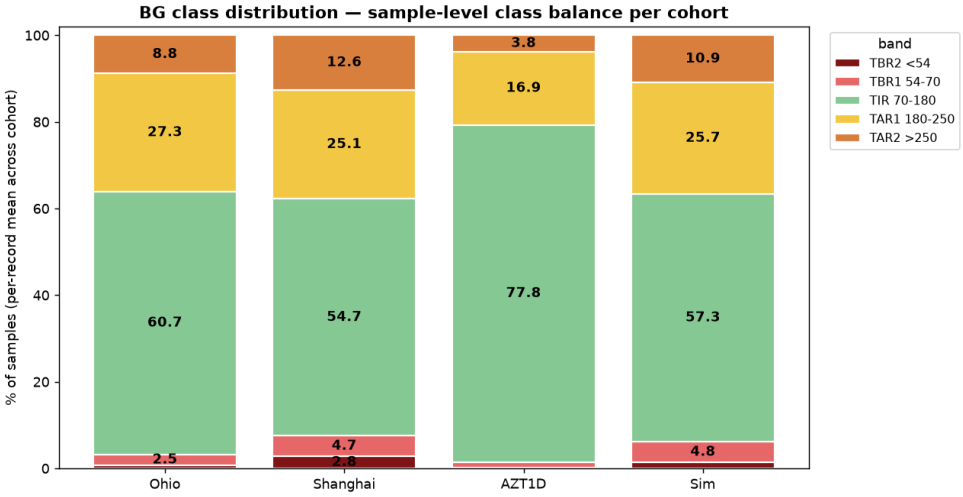


Figure 4: Per-cohort time-in-band class balance (TBR2/TBR1/TIR/TAR1/TAR2).

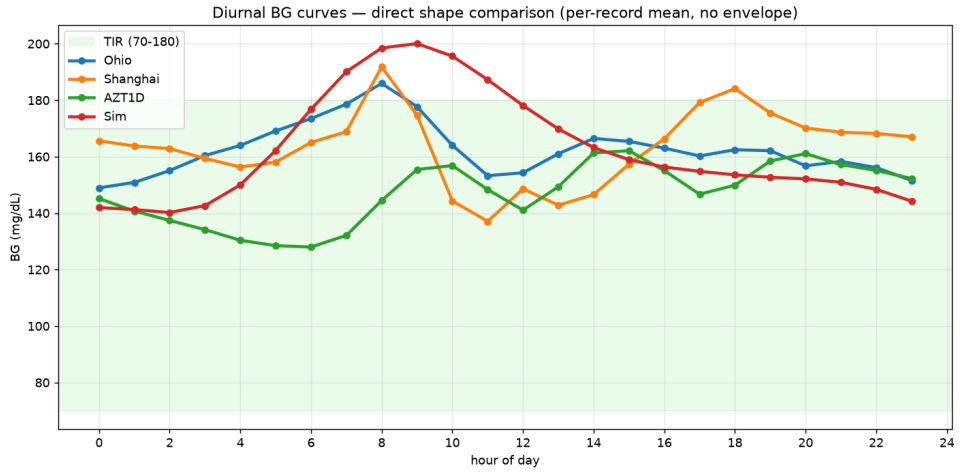


Figure 5: Hour-of-day mean blood glucose; clean line overlay isolating the diurnal shape across cohorts.

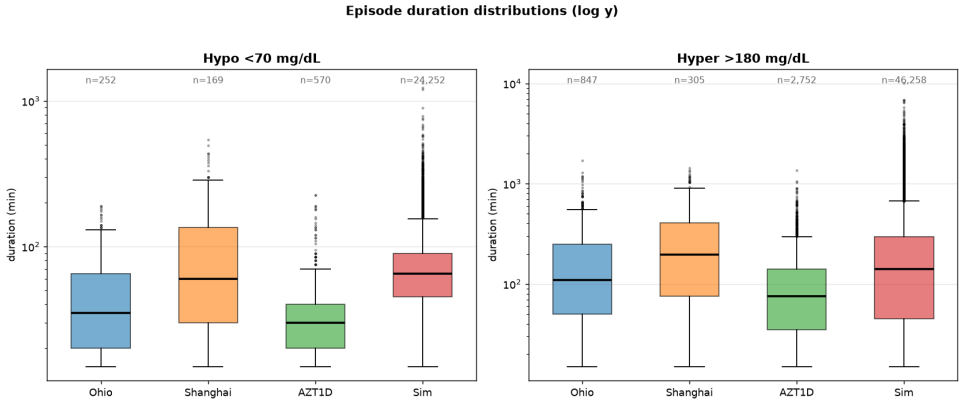


Figure 6: Hypo- and hyperglycaemic episode-duration distributions by cohort.

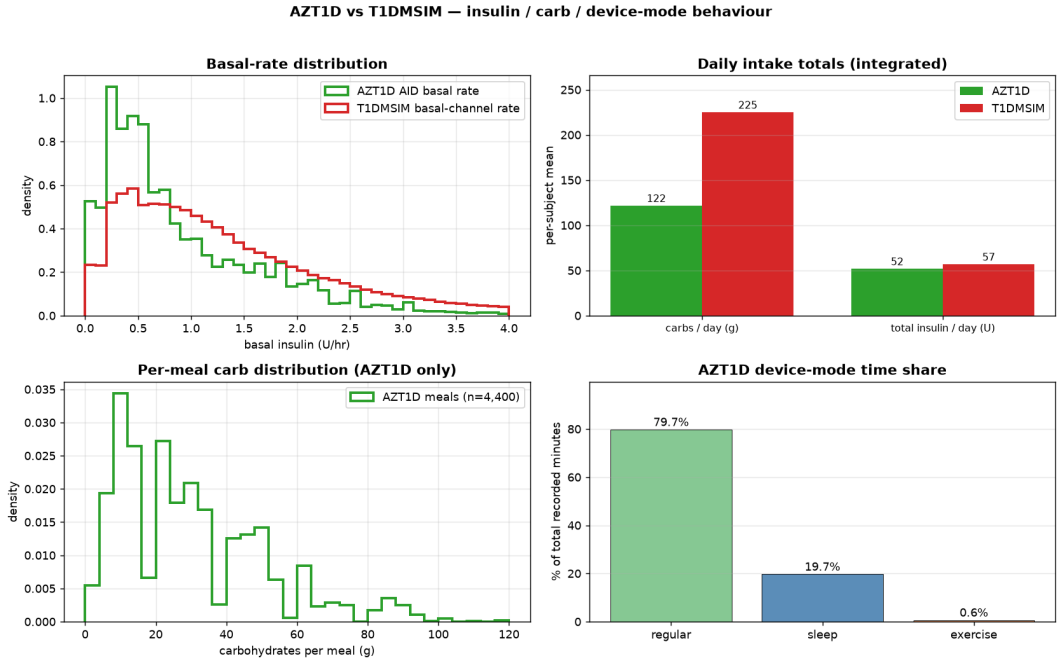


Figure 7: AZT1D versus T1DMSIM insulin/carbohydrate behaviour: basal-rate distribution and integrated daily insulin and carbohydrate totals, the channels both sources expose.

B Comparison Against the UVA/Padova Simulator

Table 6: Identical-input replay against UVA/Padova 2008 [4, 5] via the `simglucose` core [7]: pooled distributional comparison across 24 seeds (14 days each). Δ is UVA/Padova minus T1DMSIM.

Metric	T1DMSIM	UVA/Padova	Δ
Mean BG (mg/dL)	164.2	165.4	+1.2
Median BG (mg/dL)	158.7	152.8	−5.8
SD (mg/dL)	59.5	75.2	+15.8
GMI (%)	7.24	7.27	+0.03
Time $<$ 54% (TBR2)	0.93	14.13	+13.20
Time 70–180% (TIR)	58.7	44.9	−13.8
Time $>$ 250% (TAR2)	9.40	18.20	+8.80
LBGI	0.91	29.00	+28.09
HBGI	7.95	12.24	+4.29

Table 7: Self-dosed meal-excursion peak rise (mg/dL) and time-to-peak (min) by carbohydrate load; median across 774 excursions.

Meal size	n	Sim peak	UVA peak	Sim t-peak	UVA t-peak
25–40 g	355	36	24	85	70
40–55 g	205	49	35	65	75
55–70 g	104	58	44	78	80
70–200 g	110	89	53	65	80

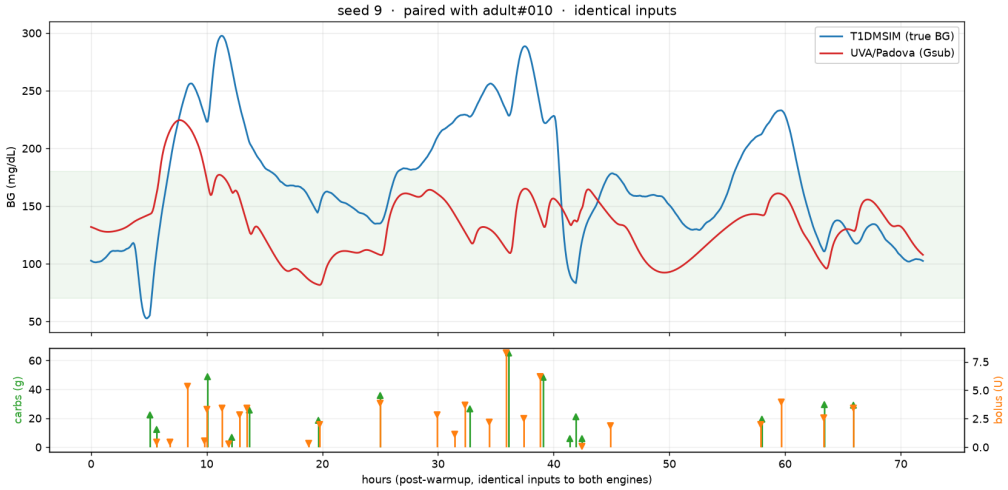


Figure 8: Representative 72-hour paired trace under identical replayed inputs: T1DMSIM versus the paired UVA/Padova patient.

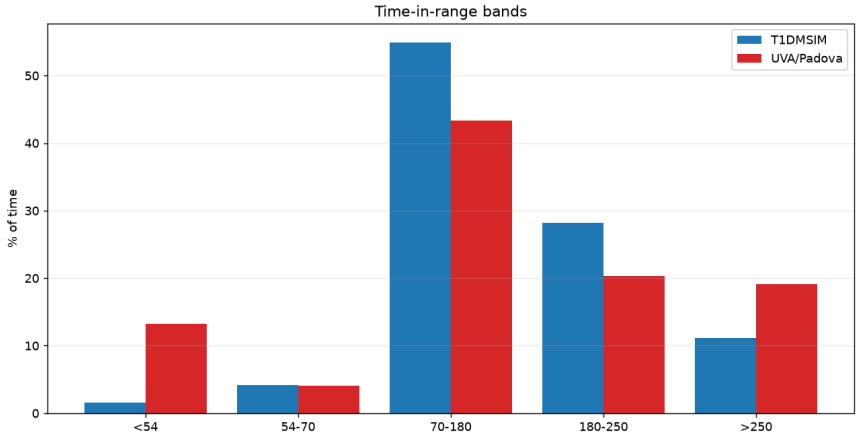


Figure 9: Time-in-range bands under identical-input replay.

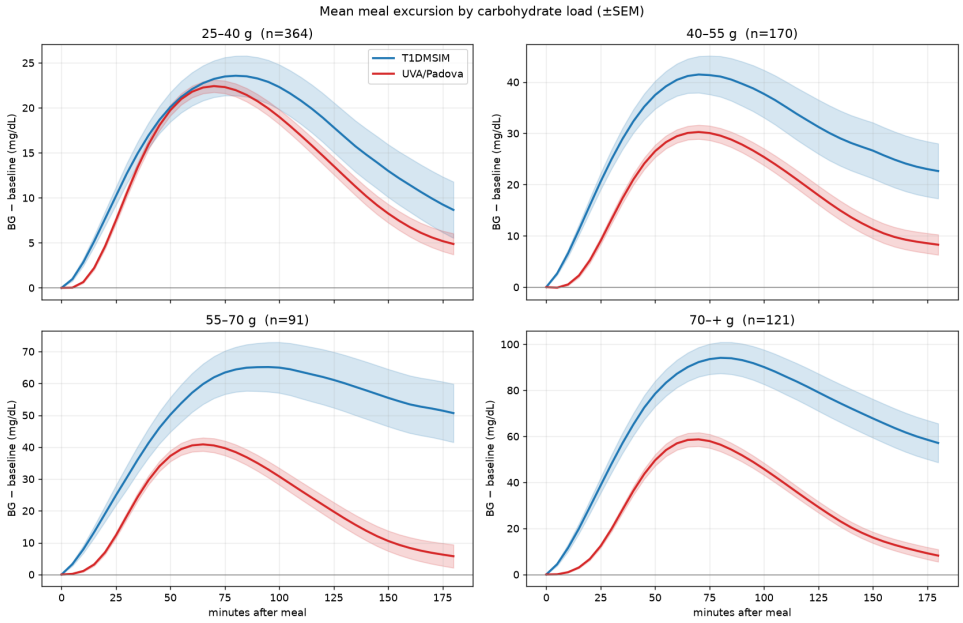


Figure 10: Mean self-dosed meal excursion by carbohydrate load for each engine.

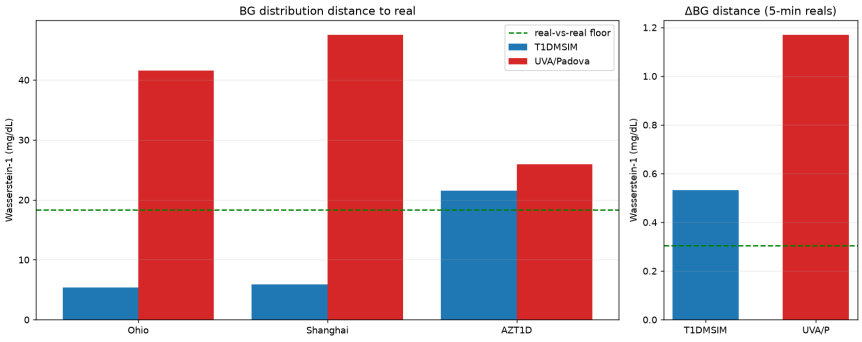


Figure 11: Wasserstein-1 distance of each simulator's pooled distribution to real CGM, with the real-vs-real floor for reference.